Lab 3 Report - Kubernetes and Minikube

Student	姜丞骏
Experiment	Lab 3: Kubernetes tutorials with Minikube
Execution	GitHub Actions Ubuntu runner with Docker, minikube, kubectl and istioctl
Evidence repository	https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio
Successful Actions run	https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio/actions/runs/27429016540
Evidence log	D:\论文\期末作业\计算机\Lab3_Lab4_Kubernetes_Istio\evidence\lab3-lab4-evidence\lab3-lab4-run.log

Objective

Follow the Kubernetes Hello Minikube and Kubernetes Basics tutorials. The experiment starts a minikube cluster, deploys applications, explores pods and logs, exposes a NodePort service, scales the deployment, performs a rolling update, and rolls the deployment back.

Major Steps and Results

1. Started minikube on the Docker driver and verified cluster-info and node status. 2. Created hello-minikube and exposed it through NodePort; curl returned the echo server response. 3. Deployed kubernetes-bootcamp, inspected pod details, logs, and environment variables. 4. Exposed bootcamp as NodePort and verified HTTP access. 5. Scaled bootcamp to 4 replicas and observed requests served by multiple pods. 6. Updated the deployment image to nginx:1.27-alpine, confirmed successful rollout, then rolled back to gcr.io/google-samples/kubernetes-bootcamp:v1.

Screenshot 1. Hello Minikube deployment, NodePort exposure, and curl verification.

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Lab 3 - Hello Minikube
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## kubectl create deployment hello-minikube --image=registry.k8s.io/echoserver:1.10
deployment.apps/hello-minikube created

## kubectl expose deployment hello-minikube --type=NodePort --port=8080
service/hello-minikube exposed

## kubectl wait --for=condition=available --timeout=180s deployment/hello-minikube
deployment.apps/hello-minikube condition met
hello-minikube URL: http://192.168.49.2:32338


Hostname: hello-minikube-6fcb694474-xrcxt

Pod Information:
	-no pod information available-

Server values:
	server_version=nginx: 1.13.3 - lua: 10008

Request Information:
	client_address=10.244.0.1
	method=GET
	real path=/
	query=
	request_version=1.1
	request_scheme=http
	request_uri=http://192.168.49.2:8080/

Request Headers:
	accept=*/*
	host=192.168.49.2:32338
	user-agent=curl/8.5.0

Request Body:
	-no body in request-


## kubectl get deployments,pods,services -o wide
NAME                             READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES
  SELECTOR
deployment.apps/hello-minikube   1/1     1            1           8s    echoserver
  registry.k8s.io/echoserver:1.10   app=hello-minikube

NAME                                  READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED
  NODE   READINESS GATES
pod/hello-minikube-6fcb694474-xrcxt   1/1     Running   0          3s    10.244.0.3   minikube   <none>
  <none>

NAME                     TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE   SELECTOR
...
```

Screenshot 2. Bootcamp scaling, rolling update, and rollback evidence.

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Lab 3 - Scale
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## kubectl scale deployments/kubernetes-bootcamp --replicas=4
deployment.apps/kubernetes-bootcamp scaled

## kubectl wait --for=condition=available --timeout=180s deployment/kubernetes-bootcamp
deployment.apps/kubernetes-bootcamp condition met

## kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
hello-minikube        1/1     1            1           20s
kubernetes-bootcamp   4/4     4            4           12s

## kubectl get pods -o wide
NAME                                   READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED
  NODE   READINESS GATES
hello-minikube-6fcb694474-xrcxt        1/1     Running   0          15s   10.244.0.3   minikube   <none>
  <none>
kubernetes-bootcamp-67fbdd6b79-5dcwx   1/1     Running   0          1s    10.244.0.5   minikube   <none>
  <none>
kubernetes-bootcamp-67fbdd6b79-jsbth   1/1     Running   0          12s   10.244.0.4   minikube   <none>
  <none>
kubernetes-bootcamp-67fbdd6b79-nhtjt   1/1     Running   0          1s    10.244.0.7   minikube   <none>
  <none>
kubernetes-bootcamp-67fbdd6b79-zxh5s   1/1     Running   0          1s    10.244.0.6   minikube   <none>
  <none>
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-jsbth | v=1
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-5dcwx | v=1
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-jsbth | v=1
Hello Kubernetes bootcamp! | Running on: kubernetes-bootcamp-67fbdd6b79-5dcwx | v=1

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Lab 3 - Update
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## kubectl scale deployments/kubernetes-bootcamp --replicas=2
deployment.apps/kubernetes-bootcamp scaled

## kubectl wait --for=condition=available --timeout=180s deployment/kubernetes-bootcamp
deployment.apps/kubernetes-bootcamp condition met

## kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=nginx:1.27-alpine
deployment.apps/kubernetes-bootcamp image updated

## kubectl rollout status deployments/kubernetes-bootcamp --timeout=300s
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
deployment "kubernetes-bootcamp" successfully rolled out

## kubectl describe deployments/kubernetes-bootcamp
Name:                   kubernetes-bootcamp
Namespace:              default
CreationTimestamp:      Fri, 12 Jun 2026 16:33:05 +0000
Labels:                 app=kubernetes-bootcamp
Annotations:            deployment.kubernetes.io/revision: 2
Selector:               app=kubernetes-bootcamp
Replicas:               2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
   kubernetes-bootcamp:
    Image:         nginx:1.27-alpine
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  kubernetes-bootcamp-67fbdd6b79 (0/0 replicas created)
NewReplicaSet:   kubernetes-bootcamp-7566cfdcf (2/2 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  16s   deployment-controller  Scaled up replica set kubernetes-
  bootcamp-67fbdd6b79 from 0 to 1
  Normal  ScalingReplicaSet  5s    deployment-controller  Scaled up replica set kubernetes-
  bootcamp-67fbdd6b79 from 1 to 4
...
```

Notes

The original tutorial image gcr.io/google-samples/kubernetes-bootcamp:v2 did not become ready reliably in the GitHub runner, so the rolling update proof used nginx:1.27-alpine as a stable replacement image. The Kubernetes update mechanism was still exercised through kubectl set image, rollout status, deployment describe, and rollout undo.

References

Kubernetes official tutorials: Hello Minikube; Deploy, Explore, Expose, Scale, and Update an application.

Lab 4 Report - Istio Getting Started

Student	姜丞骏
Experiment	Lab 4: Istio install, Gateway API, Bookinfo, and dashboard
Execution	GitHub Actions Ubuntu runner with Docker, minikube, kubectl and istioctl
Evidence repository	https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio
Successful Actions run	https://github.com/whatokk/cloud-labs-03-04-kubernetes-istio/actions/runs/27429016540
Evidence log	D:\论文\期末作业\计算机\Lab3_Lab4_Kubernetes_Istio\evidence\lab3-lab4-evidence\lab3-lab4-run.log

Objective

Follow the Istio Getting Started guide. The experiment installs Istio, installs Gateway API CRDs, deploys the Bookinfo sample, configures Gateway API routing, accesses the Product Page, and installs dashboard add-ons including Kiali, Prometheus, Grafana, and tracing services.

Major Steps and Results

1. Downloaded Istio 1.27.1 and installed the demo profile. 2. Verified istiod, ingress gateway, and egress gateway pods in istio-system. 3. Installed Gateway API CRDs and applied the Bookinfo Gateway/HTTPRoute. 4. Deployed Bookinfo services and confirmed all productpage, details, ratings, and reviews pods were running with Istio sidecars. 5. Accessed productpage through port-forward and verified the HTML title Simple Bookstore App. 6. Installed dashboard add-ons and verified Kiali, Prometheus, Grafana, tracing, and jaeger-collector services.

Screenshot 1. Istio download, istioctl version, demo profile installation, and namespace injection label.

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Lab 4 - Install Istio
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Downloading istio-1.27.1 from https://github.com/istio/istio/releases/download/1.27.1/istio-1.27.1-linux-
  amd64.tar.gz ...

Istio 1.27.1 download complete!

The Istio release archive has been downloaded to the istio-1.27.1 directory.

To configure the istioctl client tool for your workstation,
add the /home/runner/work/cloud-labs-03-04-kubernetes-istio/cloud-labs-03-04-kubernetes-
  istio/istio-1.27.1/bin directory to your environment path variable with:
         export PATH="$PATH:/home/runner/work/cloud-labs-03-04-kubernetes-istio/cloud-labs-03-04-kubernetes-
  istio/istio-1.27.1/bin"

Begin the Istio pre-installation check by running:
□ istioctl x precheck

Try Istio in ambient mode
□https://istio.io/latest/docs/ambient/getting-started/
Try Istio in sidecar mode
□https://istio.io/latest/docs/setup/getting-started/
Install guides for ambient mode
□https://istio.io/latest/docs/ambient/install/
Install guides for sidecar mode
□https://istio.io/latest/docs/setup/install/

Need more information? Visit https://istio.io/latest/docs/

## istioctl version --remote=false
client version: 1.27.1

## istioctl install --set profile=demo -y
        |\
        | \
        |  \
        |   \
      /||    \
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    /  ||      \
   /   ||       \
  /    ||        \
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WARNING: Istio 1.26.0 may be out of support (EOL) already: see
  https://istio.io/latest/docs/releases/supported-releases/ for supported releases

- Processing resources for Istio core.
OK Istio core installed
- Processing resources for Istiod.
- Processing resources for Istiod. Waiting for Deployment/istio-system/istiod
OK Istiod installed
- Processing resources for Egress gateways, Ingress gateways.
- Processing resources for Egress gateways, Ingress gateways. Waiting for Deployment/istio-system...
OK Ingress gateways installed
OK Egress gateways installed
- Pruning removed resources
...
```

Screenshot 2. Bookinfo deployment, Gateway/HTTPRoute creation, and product page verification.

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Lab 4 - Bookinfo
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## kubectl apply -f istio-1.27.1/samples/bookinfo/platform/kube/bookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created

## kubectl wait --for=condition=available --timeout=300s deployment/productpage-v1
deployment.apps/productpage-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/details-v1
deployment.apps/details-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/ratings-v1
deployment.apps/ratings-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/reviews-v1
deployment.apps/reviews-v1 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/reviews-v2
deployment.apps/reviews-v2 condition met

## kubectl wait --for=condition=available --timeout=300s deployment/reviews-v3
deployment.apps/reviews-v3 condition met

## kubectl get pods,services -o wide
NAME                                       READY   STATUS    RESTARTS   AGE   IP            NODE
  NOMINATED NODE   READINESS GATES
pod/details-v1-6cc9f5cc44-wxzdp            2/2     Running   0          24s   10.244.0.15   minikube
  <none>           <none>
pod/hello-minikube-6fcb694474-xrcxt        1/1     Running   0          64s   10.244.0.3    minikube
  <none>           <none>
pod/kubernetes-bootcamp-67fbdd6b79-l586m   1/1     Running   0          44s   10.244.0.11   minikube
  <none>           <none>
pod/kubernetes-bootcamp-67fbdd6b79-pjx5d   1/1     Running   0          45s   10.244.0.10   minikube
  <none>           <none>
pod/productpage-v1-7f885b46fc-6lvng        2/2     Running   0          24s   10.244.0.20   minikube
  <none>           <none>
pod/ratings-v1-77b8b6df5b-fr9xd            2/2     Running   0          24s   10.244.0.17   minikube
  <none>           <none>
pod/reviews-v1-fdbf79cd8-b7fvg             2/2     Running   0          24s   10.244.0.16   minikube
  <none>           <none>
pod/reviews-v2-674c6d8b4-4hrng             2/2     Running   0          24s   10.244.0.18   minikube
  <none>           <none>
pod/reviews-v3-7b775c7568-x5wq5            2/2     Running   0          24s   10.244.0.19   minikube
  <none>           <none>

NAME                          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE   SELECTOR
service/details               ClusterIP   10.111.10.90     <none>        9080/TCP         24s   app=details
service/hello-minikube        NodePort    10.107.197.131   <none>        8080:32338/TCP   69s   app=hello-
  minikube
service/kubernetes            ClusterIP   10.96.0.1        <none>        443/TCP          71s   <none>
service/kubernetes-bootcamp   NodePort    10.98.194.50     <none>        8080:30191/TCP   54s
  app=kubernetes-bootcamp
service/productpage           ClusterIP   10.109.222.249   <none>        9080/TCP         24s
  app=productpage
service/ratings               ClusterIP   10.106.151.41    <none>        9080/TCP         24s   app=ratings
service/reviews               ClusterIP   10.105.132.66    <none>        9080/TCP         24s   app=reviews

## kubectl apply -f istio-1.27.1/samples/bookinfo/gateway-api/bookinfo-gateway.yaml
gateway.gateway.networking.k8s.io/bookinfo-gateway created
httproute.gateway.networking.k8s.io/bookinfo created
error: timed out waiting for the condition on gateways/bookinfo-gateway

## kubectl get gateway,httproute -o wide
NAME                                                CLASS   ADDRESS   PROGRAMMED   AGE
gateway.gateway.networking.k8s.io/bookinfo-gateway  istio             False        60s

NAME                                          HOSTNAMES   AGE
httproute.gateway.networking.k8s.io/bookinfo              60s

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Lab 4 - Access Bookinfo
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Forwarding from 0.0.0.0:9080 -> 9080
Handling connection for 9080
Bookinfo productpage title:
<title>Simple Bookstore App</title>

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```

Screenshot 3. Dashboard add-ons and service verification.

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Lab 4 - Dashboard Add-ons
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## kubectl apply -f istio-1.27.1/samples/addons
serviceaccount/grafana created
configmap/grafana created
service/grafana created
deployment.apps/grafana created
configmap/istio-grafana-dashboards created
configmap/istio-services-grafana-dashboards created
deployment.apps/jaeger created
service/tracing created
service/zipkin created
service/jaeger-collector created
serviceaccount/kiali created
configmap/kiali created
clusterrole.rbac.authorization.k8s.io/kiali created
clusterrolebinding.rbac.authorization.k8s.io/kiali created
service/kiali created
deployment.apps/kiali created
serviceaccount/loki created
configmap/loki created
configmap/loki-runtime created
clusterrole.rbac.authorization.k8s.io/loki-clusterrole created
clusterrolebinding.rbac.authorization.k8s.io/loki-clusterrolebinding created
service/loki-memberlist created
service/loki-headless created
service/loki created
statefulset.apps/loki created
serviceaccount/prometheus created
configmap/prometheus created
clusterrole.rbac.authorization.k8s.io/prometheus created
clusterrolebinding.rbac.authorization.k8s.io/prometheus created
service/prometheus created
deployment.apps/prometheus created

## kubectl get pods -n istio-system -o wide
NAME                                    READY   STATUS              RESTARTS   AGE     IP            NODE
  NOMINATED NODE   READINESS GATES
grafana-665dcbcb9b-4vmgk                0/1     Running             0          20s     10.244.0.22
  minikube   <none>           <none>
istio-egressgateway-5fbf54b4bd-p82n9    1/1     Running             0          2m4s    10.244.0.14
  minikube   <none>           <none>
istio-ingressgateway-77c8b58566-v282r   1/1     Running             0          2m4s    10.244.0.13
  minikube   <none>           <none>
istiod-5dc9d7b78-dqts5                  1/1     Running             0          2m10s   10.244.0.12
  minikube   <none>           <none>
jaeger-75846c549c-c55tj                 1/1     Running             0          20s     10.244.0.23
  minikube   <none>           <none>
kiali-7d8fcc45dd-6zw5x                  0/1     Running             0          20s     10.244.0.24
  minikube   <none>           <none>
loki-0                                  0/2     ContainerCreating   0          20s     <none>
  minikube   <none>           <none>
prometheus-759d7bcd86-f9xpp             0/2     ContainerCreating   0          20s     <none>
  minikube   <none>           <none>

## kubectl rollout status deployment/kiali -n istio-system --timeout=240s
Waiting for deployment "kiali" rollout to finish: 0 of 1 updated replicas are available...
deployment "kiali" successfully rolled out

## kubectl rollout status deployment/prometheus -n istio-system --timeout=240s
deployment "prometheus" successfully rolled out

## kubectl get svc -n istio-system kiali prometheus grafana tracing jaeger-collector -o wide
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)
  AGE   SELECTOR
kiali              ClusterIP   10.96.217.242    <none>        20001/TCP,9090/TCP
  51s   app.kubernetes.io/instance=kiali,app.kubernetes.io/name=kiali
prometheus         ClusterIP   10.103.67.108    <none>        9090/TCP
  51s
  app.kubernetes.io/component=server,app.kubernetes.io/instance=prometheus,app.kubernetes.io/name=prometheus
grafana            ClusterIP   10.102.216.210   <none>        3000/TCP
  51s   app.kubernetes.io/instance=grafana,app.kubernetes.io/name=grafana
tracing            ClusterIP   10.105.151.84    <none>        80/TCP,16685/TCP
  51s   app=jaeger
jaeger-collector   ClusterIP   10.105.60.249    <none>        14268/TCP,14250/TCP,9411/TCP,4317/TCP,4318/TCP
  51s   app=jaeger

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```

Screenshot 4. Final cluster state and Istio proxy status.

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Final State
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## kubectl get all -A
NAMESPACE      NAME                                       READY   STATUS    RESTARTS   AGE
default        pod/bookinfo-gateway-istio-b6f96bf84-f9jzk   1/1     Running   0          2m
default        pod/details-v1-6cc9f5cc44-wxzdp              2/2     Running   0          2m25s
default        pod/hello-minikube-6fcb694474-xrcxt          1/1     Running   0          3m5s
default        pod/kubernetes-bootcamp-67fbdd6b79-l586m     1/1     Running   0          2m45s
default        pod/kubernetes-bootcamp-67fbdd6b79-pjx5d     1/1     Running   0          2m46s
default        pod/productpage-v1-7f885b46fc-6lvng          2/2     Running   0          2m25s
default        pod/ratings-v1-77b8b6df5b-fr9xd              2/2     Running   0          2m25s
default        pod/reviews-v1-fdbf79cd8-b7fvg               2/2     Running   0          2m25s
default        pod/reviews-v2-674c6d8b4-4hrng               2/2     Running   0          2m25s
default        pod/reviews-v3-7b775c7568-x5wq5              2/2     Running   0          2m25s
istio-system   pod/grafana-665dcbcb9b-4vmgk                 1/1     Running   0          51s
istio-system   pod/istio-egressgateway-5fbf54b4bd-p82n9     1/1     Running   0          2m35s
istio-system   pod/istio-ingressgateway-77c8b58566-v282r    1/1     Running   0          2m35s
istio-system   pod/istiod-5dc9d7b78-dqts5                   1/1     Running   0          2m41s
istio-system   pod/jaeger-75846c549c-c55tj                  1/1     Running   0          51s
istio-system   pod/kiali-7d8fcc45dd-6zw5x                   1/1     Running   0          51s
istio-system   pod/loki-0                                   1/2     Running   0          51s
istio-system   pod/prometheus-759d7bcd86-f9xpp              2/2     Running   0          51s
kube-system    pod/coredns-7d764666f9-b9ttq                 1/1     Running   0          3m5s
kube-system    pod/etcd-minikube                            1/1     Running   0          3m11s
kube-system    pod/kube-apiserver-minikube                  1/1     Running   0          3m11s
kube-system    pod/kube-controller-manager-minikube         1/1     Running   0          3m11s
kube-system    pod/kube-proxy-z9wt6                         1/1     Running   0          3m6s
kube-system    pod/kube-scheduler-minikube                  1/1     Running   0          3m11s
kube-system    pod/storage-provisioner                      1/1     Running   0          3m10s

NAMESPACE      NAME                                 TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)
  AGE
default        service/bookinfo-gateway-istio       LoadBalancer   10.103.22.194    <pending>
  15021:31726/TCP,80:32135/TCP                                                     2m
default        service/details                      ClusterIP      10.111.10.90     <none>        9080/TCP
  2m25s
default        service/hello-minikube               NodePort       10.107.197.131   <none>        8080:32338/TCP
  3m10s
default        service/kubernetes                   ClusterIP      10.96.0.1        <none>        443/TCP
  3m12s
default        service/kubernetes-bootcamp          NodePort       10.98.194.50     <none>        8080:30191/TCP
  2m55s
default        service/productpage                  ClusterIP      10.109.222.249   <none>        9080/TCP
  2m25s
default        service/ratings                      ClusterIP      10.106.151.41    <none>        9080/TCP
  2m25s
default        service/reviews                      ClusterIP      10.105.132.66    <none>        9080/TCP
  2m25s
istio-system   service/grafana                      ClusterIP      10.102.216.210   <none>        3000/TCP
  51s
istio-system   service/istio-egressgateway          ClusterIP      10.110.12.8      <none>        80/TCP,443/TCP
  2m35s
istio-system   service/istio-ingressgateway         LoadBalancer   10.100.32.26     <pending>
  15021:32667/TCP,80:31641/TCP,443:32665/TCP,31400:30435/TCP,15443:31213/TCP   2m35s
istio-system   service/istiod                       ClusterIP      10.110.66.182    <none>
  15010/TCP,15012/TCP,443/TCP,15014/TCP                                            2m41s
istio-system   service/jaeger-collector             ClusterIP      10.105.60.249    <none>
  14268/TCP,14250/TCP,9411/TCP,4317/TCP,4318/TCP                                   51s
istio-system   service/kiali                        ClusterIP      10.96.217.242    <none>
  20001/TCP,9090/TCP                                                               51s
istio-system   service/loki                         ClusterIP      10.107.148.161   <none>
  3100/TCP,9095/TCP                                                                51s
istio-system   service/loki-headless                ClusterIP      None             <none>        3100/TCP
  51s
istio-system   service/loki-memberlist              ClusterIP      None             <none>        7946/TCP
  51s
istio-system   service/prometheus                   ClusterIP      10.103.67.108    <none>        9090/TCP
  51s
istio-system   service/tracing                      ClusterIP      10.105.151.84    <none>
  80/TCP,16685/TCP                                                                 51s
istio-system   service/zipkin                       ClusterIP      10.98.157.122    <none>        9411/TCP
  51s
kube-system    service/kube-dns                     ClusterIP      10.96.0.10       <none>
  53/UDP,53/TCP,9153/TCP                                                           3m11s

NAMESPACE     NAME                        DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR
  AGE
```

Notes

Gateway API resources were created successfully. In the GitHub Actions minikube environment the gateway Programmed condition stayed False because no external load balancer address was assigned. The application access test therefore used kubectl port-forward to the productpage service, which returned the expected Bookinfo HTML page.

References

Istio official Getting Started guide: install, Gateway API, Bookinfo, IP/access, and dashboard sections.